

DEXTER FERNANDES

Computer Vision Engineer Bristol, UK — open to relocating to the Netherlands +44 7436 007369 dexterfernandes11@gmail.com
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SUMMARY

Computer vision engineer with 3+ years building and deploying industrial vision systems that run 24/7 on edge hardware. Delivered visual inspection, defect detection and safety perception for clients in oil and gas, construction and manufacturing, owning projects from camera tuning and dataset creation through training, quantisation, deployment on industrial PCs and on-site validation. Led a team of four engineers and worked directly with operators and customer stakeholders. MSc Robotics (Merit), University of Bristol.

RELEVANT EXPERIENCE

Esbaar — AI Engineer

01/2023 – 08/2023 · MUSCAT, OMAN

- Owned the end-to-end ML lifecycle for real-time perception on BP drilling rigs, from data curation and training through deployment and on-site validation, improving F1 by 15% under 24/7 operation.
- Developed and maintained NVIDIA DeepStream 6.x and FFmpeg video pipelines covering camera ingestion, inference and field diagnostics in desert conditions.
- Adapted YOLOv5 detectors in PyTorch, replacing backbone and neck layers and retraining for rig-specific classes, with Optuna sweeps and Weights & Biases tracking across runs.
- Benchmarked YOLOv5 against EfficientDet on a repeatable harness, quantifying accuracy, latency and stability pre and post optimisation to catch regressions before release.
- Applied image preprocessing and artefact detection with OpenCV and scikit-image to catch mud and sand contamination on camera feeds, reducing false positives by 13%.
- Defined a repeatable baseline to FP16 to validation workflow with acceptance thresholds, giving consistent latency, memory and stability outcomes across deployments.
- Built ingestion and validation pipelines across image, video and sensor data with Airflow, MLflow, DVC and CVAT, processing 3.6 million records and cutting rollout from 72h to under 24h.
- Hardened deployed services with Dockerised microservices, Kafka messaging, health checks and safe restarts, cutting live video pipeline crashes by 67% and enabling recovery without site visits.
- Led on-site validation on rig deployments, diagnosing model failures under lighting, motion and environmental stressors and turning findings into fixes for live operation.
- Presented performance findings to leadership, project managers and rig operators, turning production signals into rollout and optimisation decisions.

LivNSense Technologies — Lead Software Engineer AI

05/2020 – 12/2022 · BENGALURU, INDIA

- Built YOLOv5 visual quality inspection for John Deere engine paintjobs on a conveyor line, detecting paint defects, contamination and colour inconsistency in production.
- Led four engineers building a two-stage detector feeding helmet-colour classification for construction safety in Texas, cutting false positives by 33% within latency and power budgets.
- Built asynchronous C++17 perception pipelines on OpenVINO across Intel Movidius VPUs and Core i7 industrial edge boxes (VTC7252-7C4IP), raising throughput from 5 to 15 FPS under real-time constraints.
- Optimised architectures with quantisation-aware training, structured pruning and layer fusion in Keras and OpenVINO, cutting model size by 50% while holding production accuracy.
- Deployed quantised models with FP16 and INT8, benchmarking accuracy, latency and memory trade-offs against real-time constraints on Intel edge targets.
- Tuned camera and ISP settings to keep models reliable on RGB and IR streams across 24/7 operation, cutting false positives on IR and reducing manual recalibration.
- Built a real-time multi-camera perception pipeline on a moving construction vehicle, synchronising streams and running incident detection under edge compute constraints.
- Evaluated DeepLabv3+, Mask R-CNN, U-Net and SegFormer for annotation review, foreground separation and failure analysis on industrial camera feeds.
- Automated labelling by repurposing existing detection models and added dataset versioning with DVC, cutting manual labelling effort by 80% on large real-world datasets.
- Evaluated edge against cloud inference to minimise bandwidth, latency and operational cost while holding safety-critical performance targets.
- Set experimentation standards, reviewed model performance and mentored junior engineers, guiding optimisation strategy for production deployments.

PROJECTS

DeepStream RTSP Pipeline github.com/Dexter-Fernandes/deepstream-rtsp-pipeline

06/2026

- Built an end-to-end DeepStream 9.0 and GStreamer RTSP pipeline running YOLO detection and NvDCF tracking across 3 concurrent feeds, with metadata extraction, anonymisation and re-streamed output.
- Converted YOLO26n to a dynamic-batch TensorRT FP16 engine on 6GB VRAM Turing hardware, validating FP16 against FP32 accuracy and profiling p99 latency to sustain 29.7 FPS per stream live.
- Implemented a C++ and CUDA TensorRT decode plugin with IPluginV2DynamicExt, moving post-processing into the inference engine to support custom-layer deployment.

Modern Edge CV Benchmarking github.com/Dexter-Fernandes/ModernEdge-CV-Benchmarking

03/2026

- Benchmarked YOLOv8, YOLOv10, YOLO26 and RT-DETR on accuracy, latency and model size to compare deployment suitability for edge targets.
- Extended the comparison to EfficientViT, MobileSAM and Depth Anything v2, exporting to ONNX and testing OpenVINO deployment paths to support model selection for low-latency systems.

Presentation-Attack Detection under Generative AI

06/2026 – PRESENT

- Building an anomaly detection pipeline that combines classical texture and frequency-domain features with a fine-tuned CNN, comparing where each approach outperforms the other.
- Evaluating with the ISO/IEC 30107-3 metric set under cross-dataset protocols, with Grad-CAM for failure-mode interpretability and ONNX and TensorRT edge deployment for latency profiling.

EDUCATION

University of Bristol — MSc Robotics, Merit

09/2023 – 09/2024 · BRISTOL, ENGLAND

Dissertation: Radiological SLAM with LiDAR Odometry

- Built a real-time SLAM system in C++17 and ROS2 on a Clearpath Husky, fusing Velodyne VLP-16 LiDAR, RealSense D435i depth and radiological sensing for inspection mapping.
- Calibrated LiDAR extrinsics to the robot base and resolved transform timing with tf2 across a PCL point cloud pipeline, validating sensor placement for coverage.
- Ran the full SLAM and perception stack on an NVIDIA Jetson AGX Xavier, working within the platform's compute and memory limits for on-robot operation.

Visvesvaraya National Institute of Technology — BTech Electrical and Electronics Engineering

07/2016 – 05/2020 · NAGPUR, INDIA

- Foundation in signals, systems, embedded hardware and control, which later supported work in real-time vision, sensor systems and robotics software.

SKILLS

Languages	Python, C++17, CUDA C++, Bash, SQL
Machine vision and deep learning	object detection, segmentation, classification, anomaly and defect detection, object tracking, optical flow, YOLO (v5/v8/v10/v26), RT-DETR, EfficientDet, DeepLabv3+, Mask R-CNN, U-Net, SegFormer, EfficientViT, PyTorch, TensorFlow, Keras, OpenCV, scikit-image, NumPy
Imaging and classical vision	industrial camera systems, RGB and IR streams, ISP and camera tuning, camera and sensor calibration, image preprocessing and correction, artefact and contamination detection, multi-camera synchronisation, RealSense depth, Velodyne LiDAR
Edge deployment and optimisation	OpenVINO, TensorRT (custom plugins, dynamic batching), ONNX, quantisation-aware training, post-training quantisation, FP16/INT8, structured pruning, layer fusion, latency and throughput profiling, Intel VTune, Intel Movidius Myriad VPU, Intel Core i7 industrial edge PCs, NVIDIA Jetson AGX Xavier
Video pipelines	NVIDIA DeepStream 6.x/9.0, GStreamer, FFmpeg, RTSP, ByteTrack, NvDCF, NvSORT, multi-stream video analytics
Data and experimentation	CVAT, DVC, MLflow, Airflow, Optuna, Weights & Biases, automated labelling, dataset versioning, synthetic data generation, benchmarking harnesses, ablation studies, TIDE error analysis, COCO format, drift tracking, Grad-CAM
Engineering and delivery	Docker, CI/CD, Git, CMake, pytest, gtest, REST API design, containerised microservices, Kafka, PostgreSQL, Azure, Linux, Agile and Scrum, Jira
Robotics	ROS2, rclcpp, tf2, Gazebo, LiDAR SLAM (RTAB-Map, LIO-SAM, Cartographer, GLIM), Nav2, PCL, Open3D, ICP, sensor fusion
Professional	technical leadership, mentoring, customer-facing delivery, stakeholder communication, cross-functional project ownership

ADDITIONAL EXPERIENCE

GrowthStage — Technical Assessor, Computer Vision and Software Engineering

05/2026 – PRESENT · REMOTE

Assesses candidate technical depth across perception, SLAM and deployment.

Taco Bell — Service Champion / Team Trainer

09/2024 – PRESENT · BRISTOL, ENGLAND

Progressed to Team Trainer in a customer-facing environment, onboarding new staff and handling escalations.

CERTIFICATIONS AND AWARDS

Deep Learning Specialisation, deeplearning.ai (07/2019) Best Speaker award, ABC Toastmasters Club (01/2020)